Stage + Master thesis project proposal

Stage and thesis must be two distinct activities, formally.

You need to follow the specific procedures and calendars with deadlines for both activities, separately.

Stage

"Retrieval and organization of paleoclimate data from public repositories"

Master thesis

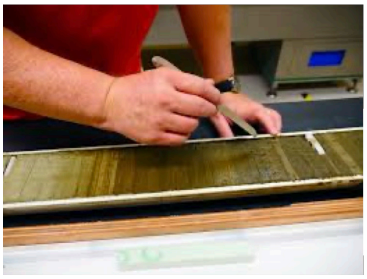
"Developing the general infrastructure to host, handle and visualize a database of past variability of mineral dust from geological archives"

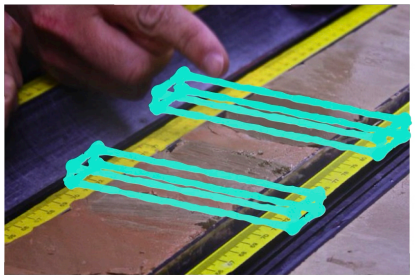
The general context of the proposed activities is that of climate sciences. The focus is on atmospheric aerosol, in particular mineral dust. Aerosols are very small (~ micrometer sized) particles that can stay aloft in the atmosphere for a few hours to several days, and be transported also for thousands of kilometers. They can have different composition and different origin, both natural and anthropogenic.

At present, we care about aerosols because of their health and air quality implications, potential damages and disruption of economic activities (e.g. dust storms), and in the context of this work especially because of their climatic impacts. In fact, aerosols interact with solar and terrestrial radiation, mediate cloud nucleation processes and atmospheric chemistry reactions, and "dark" aerosols such as carbonaceous species and mineral dust can alter snowmelt once deposited on ice- and snow-covered surfaces. In order to characterize the role of aerosols in the context of current climate change, we need to characterize aerosol variability across time scales.

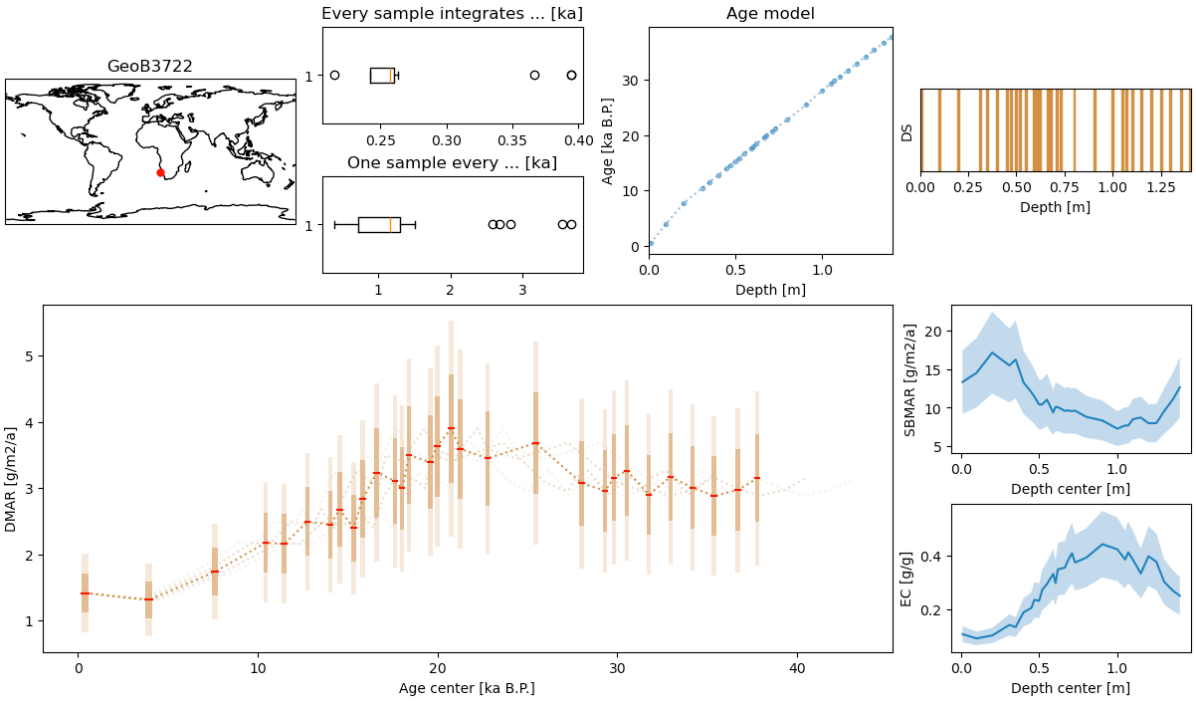
Beyond the observational era, some aerosol species such as mineral dust can be preserved in geological archives, once deposited on the surface of the planet after atmospheric transport. Examples of natural archives preserving mineral dust include ice, marine sediments, peat bogs, and loess/paleosol sequences.

The general idea is that more recent layers of aerosol deposit to the surface on top of older deposited layers. Perforations are made to retrieve sediment cores that can be subsequently analyzed, in order to obtain profiles of a given physical/chemical property (proxy) as a function of depth. When it is possible to establish a dating of the material, an age-depth model can be established. The result is a time series of past aerosol deposition.





Here is an example of visualization of a marine sediment core's Dust Mass Accumulation Rate over time with uncertainties.



More specifically, the aim of the project is to develop the general infrastructure to host, handle and visualize a database of past variability of mineral dust from geological archives.

While previous efforts in this direction exist (e.g. Albani et al., 2015; Cosentino et al., 2024), the motivation here is to expand the scope of previous work (include more data) as well as to organize the database in a more updated type of infrastructure.

Subsequent applications include e.g. climate model simulations benchmarking or inversion-based methods for data assimilation (Albani et al., 2015; Lambert et al., 2015; Molina Catricheo et al., 2024).

To enable robust model benchmarking and inversion studies, paleo dust datasets must be published in forms that support recalculation and uncertainty propagation. This requires the availability of raw or minimally processed data, explicit uncertainty estimates (including flags for unquantifiable uncertainties), and assessments of spatial representativeness.

The first step involves compiling globally distributed proxy datasets with standardized formats and rich metadata, including age models, chronological uncertainties, analytical methods, site characteristics, and proxy measurement protocols.

The second step involves transforming raw proxy measurements into standardized metrics that improve comparability both within and across proxy types. This transformation requires accounting for site-specific depositional processes, sampling resolution, and methodological differences that may bias temporal trends.

References

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